



**The University of Azad Jammu and Kashmir,
Muzaffarabad**

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Roll No: 2024-SE-06

Course Title: Database System

Course Code: SE-2204

Lab No : 09

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Department Of Software Engineering

✓ **Joins****1. INTRODUCTION**

This lab was performed to understand SQL JOIN operations using MySQL in XAMPP. Different types of joins including INNER JOIN, LEFT JOIN, RIGHT JOIN, SELF JOIN, and multi-table joins were used to retrieve related data from multiple tables in the company schema.

2. OBJECTIVE

The objective of this lab was to practice SQL JOIN operations and retrieve related records from multiple database tables.

3. Create Database

4.

5. Company schema tables created successfully.

Table	Action	Rows	Type	Collation	Size	Overhead
☐ assignment		5	InnoDB	utf8mb4_general_ci	32.0 KIB	-
☐ department		4	InnoDB	utf8mb4_general_ci	16.0 KIB	-
☐ employee		7	InnoDB	utf8mb4_general_ci	32.0 KIB	-
☐ project		4	InnoDB	utf8mb4_general_ci	32.0 KIB	-
4 tables	Sum	20	InnoDB	utf8mb4_general_ci	112.0 KIB	0 B

☐ Check all

Sample data inserted into all company schema tables.

Joins Part B

`SELECT * FROM `assignment``

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: | Sort by key: None

Extra options

		EmpID	ProjectID	WeeklyHours
☐	Edit Copy Delete	1	1	20
☐	Edit Copy Delete	2	3	15
☐	Edit Copy Delete	3	2	18
☐	Edit Copy Delete	6	1	25
☐	Edit Copy Delete	7	4	12

☐ Check all | With selected: Edit Copy Delete Export

Showing rows 0 - 3 (4 total, Query took 0.0000 seconds.)

`SELECT * FROM `department``

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

☐ Show all | Number of rows: 25 | Filter rows: | Sort

Extra options

		DeptID	DeptName	Location
☐	Edit Copy Delete	1	Engineering	Lahore
☐	Edit Copy Delete	2	HR	Karachi
☐	Edit Copy Delete	3	Marketing	Islamabad
☐	Edit Copy Delete	4	Finance	Lahore

☐ Check all | With selected: Edit Copy Delete Export

Joins Part B

`SELECT * FROM `employee``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: | Sort by key: None

Extra options

		▼	EmpID	EmpName	Salary	City	DeptID	ManagerID
☐	Edit	Copy	Delete	1	Ali Khan	95000.00	Lahore	1
								NULL
☐	Edit	Copy	Delete	2	Sara Ahmed	72000.00	Karachi	2
								1
☐	Edit	Copy	Delete	3	Bilal Hussain	68000.00	Islamabad	3
								1
☐	Edit	Copy	Delete	4	Maham Noor	88000.00	Lahore	2
								1
☐	Edit	Copy	Delete	5	Usman Tariq	120000.00	Karachi	1
								NULL
☐	Edit	Copy	Delete	6	Areeba Khan	85000.00	Lahore	1
								5
☐	Edit	Copy	Delete	7	Daniyal Raza	76000.00	Islamabad	4
								5

`SELECT * FROM `project``

☐ Profiling [\[Edit inline \]](#) [\[Edit \]](#) [\[Explain SQL \]](#) [\[Create PHP code \]](#) [\[Refresh \]](#)

☐ Show all | Number of rows: 25 | Filter rows: | Sort by key:

Extra options

		▼	ProjectID	ProjectName	DeptID	StartDate
☐	Edit	Copy	Delete	1	Mobile App	1 2024-01-10
☐	Edit	Copy	Delete	2	Website Redesign	3 2024-03-15
☐	Edit	Copy	Delete	3	HR System	2 2023-11-20
☐	Edit	Copy	Delete	4	Finance Tracker	NULL 2024-02-05

☐ Check all | With selected: Edit Copy Delete Export

6. Part B Queries

➤ TaskB1

- This query displays each employee along with their manager's name using SELF JOIN.

```
SELECT e.EmpName AS  
EmployeeName,  
m.EmpName AS ManagerName  
FROM Employee e  
LEFT JOIN Employee m  
ON e.ManagerID = m.EmpID;
```

Showing rows 0 - 6 (7 total, Query took 0.0006 seconds.)

-- Task B1 SELECT e.EmpName AS EmployeeName, m.EmpName AS ManagerName FROM Employee e LEFT JOIN Employee m ON e.ManagerID = m.EmpID;

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

EmployeeName	ManagerName
Ali Khan	NULL
Sara Ahmed	Ali Khan
Bilal Hussain	Ali Khan
Maham Noor	Ali Khan
Usman Tariq	NULL
Areeba Khan	Usman Tariq
Daniyal Raza	Usman Tariq

Task B1 showing employees and their managers.

➤ TaskB2

- This query lists employees who earn more salary than their managers.

```
SELECT e.EmpName,  
e.Salary AS EmployeeSalary,  
m.EmpName AS ManagerName,
```

Joins Part B

```
m.Salary AS ManagerSalary
FROM Employee e
INNER JOIN Employee m
ON e.ManagerID = m.EmpID
WHERE e.Salary > m.Salary;
```



Task B2 showing employees earning more than their managers.

➤ TaskB3

- This query displays employees whose managers work in different departments.

```
SELECT e.EmpName,
m.EmpName AS ManagerName,
d1.DeptName AS
EmployeeDepartment,
d2.DeptName AS
ManagerDepartment
FROM Employee e
INNER JOIN Employee m
ON e.ManagerID = m.EmpID
INNER JOIN Department d1
ON e.DeptID = d1.DeptID
INNER JOIN Department d2
ON m.DeptID = d2.DeptID
```

Joins Part B

WHERE e.DeptID <> m.DeptID;

Showing rows 0 - 3 (4 total. Query took 0.0009 seconds.)

```
-- Task B3 SELECT e.EmpName, m.EmpName AS ManagerName, d1.DeptName AS EmployeeDepartment, d2.DeptName AS ManagerDepartment FROM Employee e INNER JOIN Employee m ON e.ManagerID = m.EmpID INNER JOIN Department d1 ON e.DeptID = d1.DeptID INNER JOIN Department d2 ON m.DeptID = d2.DeptID WHERE e.DeptID <> m.DeptID;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

EmpName	ManagerName	EmployeeDepartment	ManagerDepartment
Sara Ahmed	Ali Khan	HR	Engineering
Maham Noor	Ali Khan	HR	Engineering
Bilal Hussain	Ali Khan	Marketing	Engineering
Daniyal Raza	(Jaman Tariq	Finance	Engineering

Task B3 showing employees and managers from different departments

➤ TaskB4

- This query displays employees with project names and weekly working hours.

```
SELECT e.EmpName,  
p.ProjectName,  
a.WeeklyHours  
FROM Employee e  
INNER JOIN Assignment a  
ON e.EmpID = a.EmpID  
INNER JOIN Project p  
ON a.ProjectID = p.ProjectID;
```

Showing rows 0 - 4 (5 total. Query took 0.0006 seconds.)

```
-- Task B4 SELECT e.EmpName, p.ProjectName, a.WeeklyHours FROM Employee e INNER JOIN Assignment a ON e.EmpID = a.EmpID INNER JOIN Project p ON a.ProjectID = p.ProjectID;
```

[Edit inline] [Edit] [Create PHP code]

☐ Show all Number of rows: 25 Filter rows: Search this table Sort by key: None

Extra options

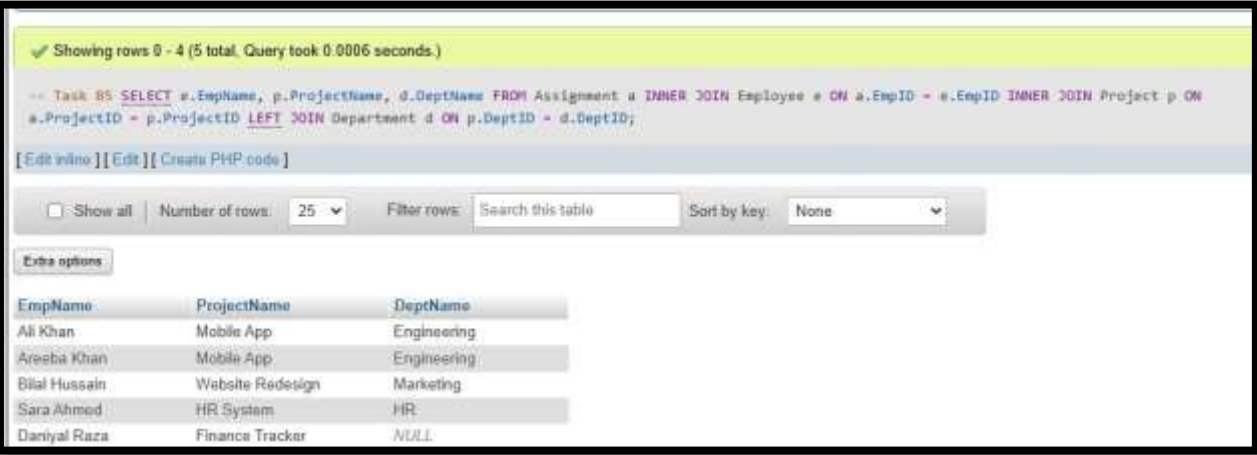
EmpName	ProjectName	WeeklyHours
Ali Khan	Mobile App	20
Arooba Khan	Mobile App	25
Bilal Hussain	Website Redesign	18
Sara Ahmed	HR System	15
Daniyal Raza	Finance Tracker	12

Task B4 showing employees with assigned projects and hours.

➤ TaskB5

- This query displays assignment details including employee name, project name, and department name.

```
SELECT e.EmpName,
p.ProjectName,
d.DeptName
FROM Assignment a
INNER JOIN Employee e
ON a.EmpID = e.EmpID
INNER JOIN Project p
ON a.ProjectID = p.ProjectID
LEFT JOIN Department d
ON p.DeptID = d.DeptID;
```



Showing rows 0 - 4 (5 total. Query took 0.0006 seconds.)

-- Task B5 SELECT e.EmpName, p.ProjectName, d.DeptName FROM Assignment a INNER JOIN Employee e ON a.EmpID = e.EmpID INNER JOIN Project p ON a.ProjectID = p.ProjectID LEFT JOIN Department d ON p.DeptID = d.DeptID;

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

EmpName	ProjectName	DeptName
Ali Khan	Mobile App	Engineering
Areeba Khan	Mobile App	Engineering
Bilal Hussain	Website Redesign	Marketing
Sara Ahmed	HR System	HR
Daniyal Raza	Finance Tracker	NULL

Task B5 showing assignment and project department details.

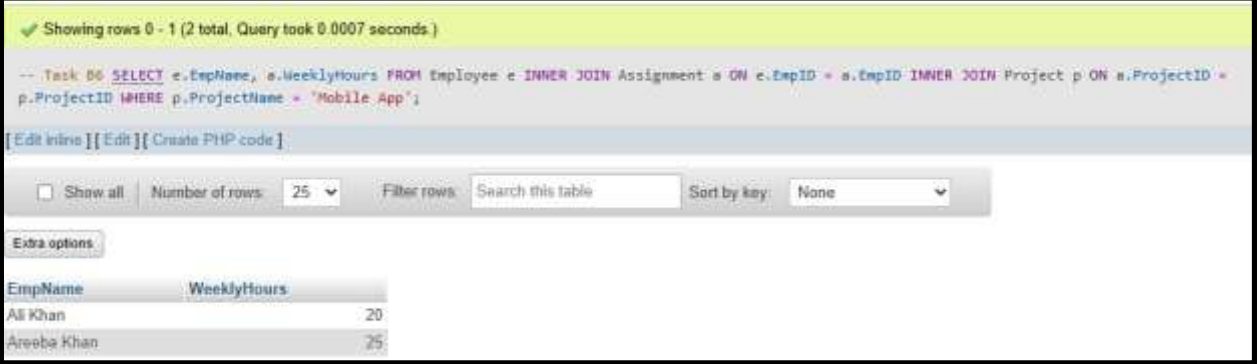
➤ TaskB6

- This query lists employees working on the Mobile App project with their weekly hours.

```
SELECT e.EmpName,
a.WeeklyHours
FROM Employee e
INNER JOIN Assignment a
ON e.EmpID = a.EmpID
INNER JOIN Project p
```

Joins Part B

```
ON a.ProjectID = p.ProjectID
WHERE p.ProjectName =
'Mobile App';
```



The screenshot shows a database query result in a web interface. At the top, a status bar indicates 'Showing rows 0 - 1 (2 total, Query took 0.0007 seconds)'. Below this, the SQL query is displayed: `-- Task B6 SELECT e.EmpName, e.WeeklyHours FROM Employee e INNER JOIN Assignment a ON e.EmpID = a.EmpID INNER JOIN Project p ON a.ProjectID = p.ProjectID WHERE p.ProjectName = 'Mobile App';`. Below the query, there are controls for 'Show all', 'Number of rows' (set to 25), 'Filter rows' (with a search box), and 'Sort by key' (set to None). An 'Extra options' button is also present. The result table has two columns: 'EmpName' and 'WeeklyHours'. It contains two rows: 'Ali Khan' with 20 hours and 'Areeba Khan' with 25 hours.

EmpName	WeeklyHours
Ali Khan	20
Areeba Khan	25

Task B6 showing employees working on the Mobile App project.

➤ TaskB7

- This query displays Lahore employees with their assigned projects and weekly hours including employees without assignments.

```
SELECT e.EmpName,
p.ProjectName,
a.WeeklyHours
FROM Employee e
LEFT JOIN Assignment a
ON e.EmpID = a.EmpID
LEFT JOIN Project p
ON a.ProjectID = p.ProjectID
WHERE e.City = 'Lahore';
```

Joins Part B

Showing rows 0 - 2 (3 total, Query took 0.0005 seconds)

-- Task B7: `SELECT e.EmpName, p.ProjectName, a.WeeklyHours FROM Employee e LEFT JOIN Assignment a ON e.EmpID = a.EmpID LEFT JOIN Project p ON a.ProjectID = p.ProjectID WHERE e.City = 'Lahore';`

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

EmpName	ProjectName	WeeklyHours
Ali Khan	Mobile App	20
Maham Noor	NULL	NULL
Areeba Khan	Mobile App	25

Task B7 showing Lahore employees and their projects.

➤ TaskB8

- This query lists employees working on projects managed by departments different from their own departments.

```
SELECT DISTINCT e.EmpName
FROM Employee e
INNER JOIN Assignment a
ON e.EmpID = a.EmpID
INNER JOIN Project p
ON a.ProjectID = p.ProjectID
WHERE e.DeptID <> p.DeptID;
```

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0007 seconds)

-- Task B8: `SELECT DISTINCT e.EmpName FROM Employee e INNER JOIN Assignment a ON e.EmpID = a.EmpID INNER JOIN Project p ON a.ProjectID = p.ProjectID WHERE e.DeptID <> p.DeptID;`

[Edit inline] [Edit] [Create PHP code]

EmpName

Query results operations

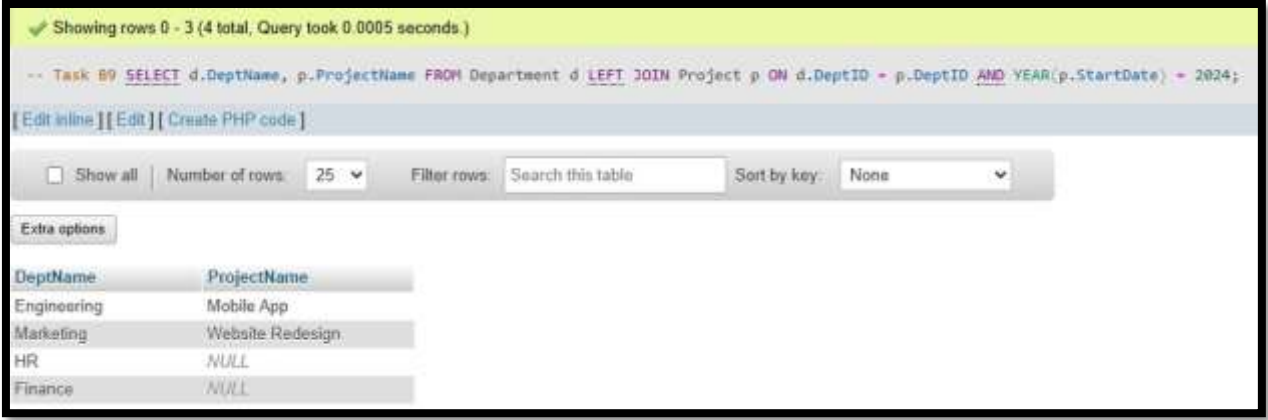
Create view

Task B8 showing employees working across different departments

➤ TaskB9

- This query displays departments and projects started in the year 2024 including departments without projects.

```
SELECT d.DeptName,  
p.ProjectName  
FROM Department d  
LEFT JOIN Project p  
ON d.DeptID = p.DeptID  
AND YEAR(p.StartDate) = 2024;
```



The screenshot shows a database query interface with a green status bar at the top indicating 'Showing rows 0 - 3 (4 total, Query took 0.0005 seconds.)'. Below this, the SQL query for Task B9 is displayed: `-- Task B9: SELECT d.DeptName, p.ProjectName FROM Department d LEFT JOIN Project p ON d.DeptID = p.DeptID AND YEAR(p.StartDate) = 2024;`. The interface includes controls for 'Show all', 'Number of rows' (set to 25), 'Filter rows' (with a search box), and 'Sort by key' (set to None). An 'Extra options' button is also present. The query results are shown in a table with two columns: 'DeptName' and 'ProjectName'. The data rows are: Engineering (Mobile App), Marketing (Website Redesign), HR (NULL), and Finance (NULL).

DeptName	ProjectName
Engineering	Mobile App
Marketing	Website Redesign
HR	NULL
Finance	NULL

Task B9 showing departments with projects started in 2024.

➤ TaskB10

- This query displays every employee with total weekly working hours across all projects including employees with zero hours.

```
SELECT e.EmpName,  
IFNULL(SUM(a.WeeklyHours),0)  
AS TotalHours  
FROM Employee e  
LEFT JOIN Assignment a  
ON e.EmpID = a.EmpID
```

Joins Part B

GROUP BY e.EmpName;



Showing rows 0 - 6 (7 total, Query took 0.0011 seconds)

-- Task B10 SELECT e.EmpName, IFNULL(SUM(a.WeeklyHours),0) AS TotalHours FROM Employee e LEFT JOIN Assignment a ON e.EmpID = a.EmpID GROUP BY e.EmpName;

[Edit inline] [Edit] [Create PHP code]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

EmpName	TotalHours
Ali Khan	20
Areeba Khan	25
Bilal Hussain	18
Daniyal Raza	12
Maham Noor	0
Sara Ahmad	15
Usman Tariq	0

Task B10 showing total weekly hours for each employee.

7. Conclusion

In this lab, different SQL JOIN operations were performed successfully using MySQL in XAMPP. INNER JOIN, LEFT JOIN, RIGHT JOIN, SELF JOIN, and multi-table joins helped in retrieving related information from multiple tables. The lab improved understanding of relational database concepts and SQL query writing.
